

TrajScale: Decoupled Mixed-Resolution Sampling for Efficient High-Resolution Video Generation

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Abstract

High-resolution video generation is costly because every denoising step processes long spatiotemporal latent sequences at the target resolution. Although early steps mainly determine global structure and motion while later steps recover detail, mixed-resolution sampling must bridge residual endpoint error, non-scale-equivariant VAE latent spaces, and a schedule-consistent high-resolution state. We introduce **TrajScale**, a decoupled mixed-resolution sampling framework that performs most denoising at low resolution and reserves a short high-resolution suffix. TrajScale decomposes the resolution transition into a LoRA-based **Endpoint Alignment Adapter (EAA)** that calibrates the current clean prediction toward the converged low-resolution endpoint, a **Cross-Resolution Latent Upsampler (CRLU)** that maps clean latents across spatial scales, and parameter-free **Target-Resolution Re-noising (TRR)** that reconstructs a schedule-consistent state for high-resolution refinement. EAA and CRLU are trained separately with supervision generated by the frozen target model, requiring no external paired videos or auxiliary teachers. On 50-step Wan2.1 with a $368 \times 640 \rightarrow 720 \times 1248$ transition, TrajScale-Q and TrajScale-E achieve $1.83\times$ and $2.22\times$ end-to-end speedups while retaining 97.76% and 97.53% of Native-HR VBench-5. Compared with trilinear interpolation, CRLU reduces latent L_1 error by 48.6% and LPIPS by 73.3%, while EAA reduces endpoint L_1 error by 25.8% and 21.0% at steps 40 and 45. TrajScale provides consistent quality-efficiency operating points across standard 50-step and 4-step distilled samplers.

1 Introduction

Recent latent video diffusion models such as Wan generate semantically coherent videos with realistic motion and fine details (Wan et al. 2025), but their inference cost grows aggressively with spatial resolution, video length, and sampling steps. In DiT-based models, every denoising iteration processes the full spatiotemporal token sequence. Executing the entire trajectory at the target resolution forces early steps—which mainly establish coarse structure and macro-motion—to bear the full burden of high-resolution computation, severely limiting real-time deployment. Since diffusion sampling inherently distributes generation across stages (global layout in early steps, high-frequency textures in later steps), a promising paradigm is mixed-resolution inference: modeling global content in a low-resolution latent space first,

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then switching to the target resolution for a short refinement suffix. This spatial-scaling approach (ModelTC 2026; Zheng et al. 2026) is fundamentally orthogonal and complementary to temporal step-distillation methods.

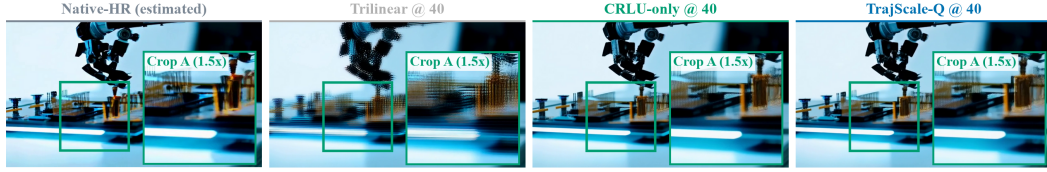
Yet, switching resolution within a diffusion trajectory is substantially more complex than resizing a latent tensor. The transition is governed by three coupled error sources: the residual gap to the converged low-resolution endpoint, the discrepancy between clean latent spaces at different resolutions, and the residual error left for high-resolution refinement. Directly regressing from the handoff state to a high-resolution endpoint forces a single Joint Trajectory-Scale Mapper (JTSM) to absorb timestep-dependent trajectory error, VAE-specific scale mismatch, and underdetermined high-frequency content. Our transition-step scans show that this entanglement attenuates fine detail, motivating a decomposition into localized subproblems.

We therefore propose **TrajScale**, a decoupled mixed-resolution sampling framework with three components activated at the resolution transition: a LoRA-based **Endpoint Alignment Adapter (EAA)** that calibrates the current clean prediction toward the converged low-resolution endpoint; a **Cross-Resolution Latent Upsampler (CRLU)** that maps clean latents across spatial scales; and **Target-Resolution Re-noising (TRR)**, a parameter-free operation that reconstructs a schedule-consistent high-resolution state for suffix refinement. EAA and CRLU are trained separately through model-endogenous self-distillation using supervision generated by the frozen target generator.

Our primary contributions are summarized as follows:

- **We propose TrajScale**, a lightweight mixed-resolution sampling framework that executes a low-resolution prefix followed by a short high-resolution suffix, yielding controllable quality-efficiency operating points (TrajScale-Q/E).
- **We decouple the resolution transition** into endpoint alignment, cross-resolution clean-latent upsampling, and target-resolution re-noising through EAA, CRLU, and TRR. The decomposition isolates trajectory, scale, and refinement errors that are entangled in joint mapping.
- **We develop model-endogenous dual supervision** without external paired videos or auxiliary teachers, and establish cached-prefix consistency for handoff-local adap-

(a) Matched real video frames — prompt 05, seed 9705, frame 41



(b)

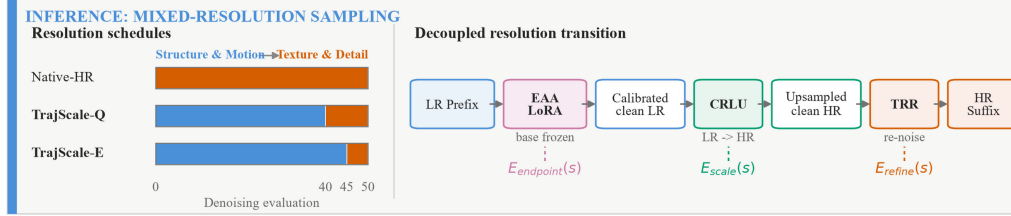


Figure 1: Real-frame comparison and TrajScale inference. (a) Content-aligned Native-HR (estimated), Trilinear, CRLU-only, and TrajScale-Q outputs for prompt 05, seed 9705, frame 41, with identical $1.5\times$ crops. The reference follows a complete 368p trajectory; transition outputs are 720p. (b) Native-HR performs all 50 evaluations at high resolution, whereas TrajScale-Q and TrajScale-E execute the first 40 and 45 evaluations at low resolution and retain 10 and 5 high-resolution suffix evaluations.

tation. Experiments on standard and distilled Wan samplers validate spatial acceleration and compatibility with few-step sampling.

2 Related Work

2.1 Latent Video Generation and Cascaded Upscaling

Latent Diffusion Models compress the generative process into a compact space for scalable high-resolution video synthesis (Rombach et al. 2022). Traditional frameworks or recent systems like LTX-2 and LUVE typically employ cascaded pipelines, using independent spatial latent upsamplers or post-generation refinement modules to elevate resolution (Blattmann et al. 2023; HaCohen et al. 2026; Zhao et al. 2026). Similarly, video super-resolution methods focus on restoring already decoded RGB outputs (Zhou et al. 2024; Xu et al. 2024).

Unlike approaches that treat upscaling as an isolated post-processing stage, TrajScale inserts CRLU directly into the frozen Wan sampling trajectory. TRR then reconstructs the target-resolution noisy state at the designated schedule level, allowing the remaining Wan steps to synthesize high-frequency content with the base model’s native generative prior.

2.2 Mixed-Resolution Diffusion Inference

The changing-resolution inference in LightX2V utilizes fixed interpolation to adjust latent sizes mid-sampling (ModelTC 2026). Concurrently, RALU demonstrates that training-free latent upscaling suffers from high-frequency aliasing and a resolution-dependent noise–timestep mismatch, designing a mixed-resolution scheme for Diffusion Transformers (Jeong et al. 2025). For image flow matching, MrFlow combines low-resolution structure generation with low-noise high-resolution refinement (Zheng et al. 2026). TrajScale extends this paradigm to video by learning cross-resolution clean-latent upsampling inside the original sampling process while

explicitly calibrating the residual low-resolution endpoint error before the resolution transition.

2.3 Few-Step Distillation and On-Policy Trajectory Learning

To accelerate inference, temporal distillation methods minimize sampling steps via trajectory regression or consistency mapping (Luo et al. 2023; Yin et al. 2024; Zhai et al. 2024). Recent works like Self-Forcing and AnyFlow further emphasize enforcing strict consistency between student training states and actual inference trajectories to mitigate distribution shift (Huang et al. 2025; Gu et al. 2026).

TrajScale focuses on spatial rather than temporal computation, making it orthogonal to step reduction. Since EAA remains inactive prior to the transition step, its parameter updates do not alter the low-resolution prefix trajectory. Leveraging this local activation property, Section 3.5 establishes the consistency conditions that allow cached prefix states to be directly accessed during inference without training–inference distribution drift.

3 Method

3.1 Problem Definition and Overall Framework

Let $v_\phi(x_s, s, c)$ denote a frozen Wan flow predictor, where x_s represents the intermediate latent before the s -th denoising calculation, c is the text condition, and T is the total number of denoising steps. We use the superscripts L and H to signify the low-resolution and high-resolution latent spaces, respectively. Let z_T^L denote the endpoint of the complete low-resolution trajectory generated by the frozen model under a given text prompt, random seed, initial noise, and low-resolution scheduler. We define (z_0^L, z_0^H) as the cross-resolution clean latent pair obtained by encoding the low- and high-resolution versions of the same video through the Wan VAE.

TrajScale introduces two independently trained modules: the Endpoint Alignment Adapter (EAA), parameterized

by handoff-specific LoRA weights $\Delta\theta_s$, and the Cross-Resolution Latent Upsampler (CRLU), denoted by U_ψ^{CRLU} with parameters ψ . Target-Resolution Re-noising (TRR) is a parameter-free state-construction operation. As illustrated in the upper panel of Figure 2, the complete transition is formulated as:

$$x_s^L \xrightarrow{\text{EAA}: v_\phi + \Delta\theta_s} \tilde{z}_s^L \xrightarrow{U_\psi^{\text{CRLU}}} \tilde{z}_s^H \xrightarrow{\text{TRR}} x_{s+1}^H.$$

Crucially, EAA only calibrates the current low-resolution clean prediction toward the converged low-resolution endpoint, whereas CRLU only learns the cross-resolution mapping between clean latent spaces. These two modules are trained separately without joint backpropagation. This decoupled training strategy prevents the massive memory overhead of hosting both the 14B denoiser and the CRLU module simultaneously in the compute graph, while enabling an isolated experimental analysis of trajectory and scale errors.

The transition step governs three distinct error sources. Let $E_{\text{endpoint}}(s)$ denote the residual discrepancy to the converged low-resolution endpoint, $E_{\text{scale}}(s)$ the cross-resolution clean-latent mapping error, and $E_{\text{refine}}(s)$ the residual high-resolution error after TRR. The total transition error can be conceptualized as:

$$E_{\text{transition}}(s) \approx E_{\text{endpoint}}(s) + E_{\text{scale}}(s) + E_{\text{refine}}(s).$$

This decomposition exposes the central trade-off in selecting s . A later transition reduces $E_{\text{endpoint}}(s)$ and moves the CRLU input closer to its clean-latent training distribution, thereby reducing $E_{\text{scale}}(s)$. However, it leaves fewer high-resolution suffix evaluations and may increase $E_{\text{refine}}(s)$. EAA, CRLU, and TRR address these three sources separately.

3.2 Model-Endogenous Dual-Supervision Data

TrajScale operates entirely without external paired videos, pre-trained super-resolution weights, or additional teacher models. The supervision for both CRLU and EAA is generated natively by the frozen target generator, tracking two distinct conditional distributions as shown in the lower panel of Figure 2.

Cross-Resolution Upsampling Pairs. We run the frozen Wan model using a set of predefined text prompts and fixed random seeds to generate high-resolution reference videos. These videos are spatially downsampled in the RGB pixel space. Both the high- and low-resolution variants are then encoded using the same Wan VAE to produce perfectly aligned cross-resolution clean latent pairs (z_0^L, z_0^H) . This construction preserves the frame count, semantic content, and motion trajectories, forcing CRLU to learn cross-scale correspondences tightly bound to the target model’s specific VAE and generative distribution.

Endpoint Alignment Pairs. Under identical text prompts, random seeds, initial noise, and schedulers, we roll out the complete low-resolution trajectory of the frozen model. We cache the intermediate low-resolution state x_s^L immediately preceding the transition step and utilize the final low-resolution trajectory endpoint z_T^L as the ground-truth target.

Consequently, CRLU receives cross-resolution structural supervision, whereas EAA receives handoff-local endpoint supervision, forming a unified model-endogenous trajectory-scale self-distillation framework.

Although both types of supervision share the same base generative model, they map to different conditional distributions: the low-resolution input for CRLU is derived from downsampled high-resolution video encodings, whereas EAA directly targets the endpoints of native low-resolution trajectories. Section 4.2 evaluates their end-to-end synergy via a 2×2 factorial design crossing trajectory states and resolution operators.

3.3 Cross-Resolution Clean-Latent Upsampling

The Cross-Resolution Latent Upsampler (CRLU) takes a 16-channel Wan VAE latent as input and produces an upscaled 16-channel latent, preserving the temporal dimension:

$$U_\psi^{\text{CRLU}} : \mathbb{R}^{B \times 16 \times F \times h \times w} \rightarrow \mathbb{R}^{B \times 16 \times F \times H \times W}.$$

The architecture adopts an encoder-resampler-decoder structure inspired by LTX-2. A 3D convolutional input layer projects the latents into a 256-dimensional hidden space, followed by 4 spatiotemporal residual blocks to extract rich spatiotemporal features. A spatial resampler then scales the resolution, after which 4 additional 3D residual blocks and a final 3D convolutional layer reconstruct the 16-channel output. This spatiotemporal processing enables the model to exploit information from adjacent latent frames while keeping the total frame count invariant.

We implement two distinct spatial resampling configurations. For the $480 \times 832 \rightarrow 720 \times 1248$ transition, the latent grid expands from 60×104 to 90×156 . The network first inflates the channel dimension by a factor of 9, applies a $3 \times$ spatial PixelShuffle, and finally performs a stride-2 down-sampling using a fixed binomial blur kernel to achieve a rational $3/2 \times$ upscaling:

$$60 \times 104 \xrightarrow{\times 3} 180 \times 312 \xrightarrow{/2} 90 \times 156.$$

For the $368 \times 640 \rightarrow 720 \times 1248$ transition, the latent grid expands from 46×80 to 90×156 . Since the target size is not an exact integer multiple, the network executes a $2 \times$ PixelShuffle to reach 92×160 , followed by a center crop to isolate the exact 90×156 target grid.

The training objective for CRLU is defined as:

$$\mathcal{L}_{\text{CRLU}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{content}} \mathcal{L}_{\text{content}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}},$$

where

$$\mathcal{L}_{\text{rec}} = \mathbb{E} \left[\sqrt{(\hat{z}_0^H - z_0^H)^2 + \epsilon^2} \right].$$

$$\mathcal{L}_{\text{content}} = \|D(\hat{z}_0^H) - z_0^L\|_1, \quad (1)$$

$$\mathcal{L}_{\text{temp}} = \|\Delta_F \hat{z}_0^H - \Delta_F z_0^H\|_1. \quad (2)$$

Here, D denotes a downscaling operator that maps the predicted high-resolution latent back to the low-resolution

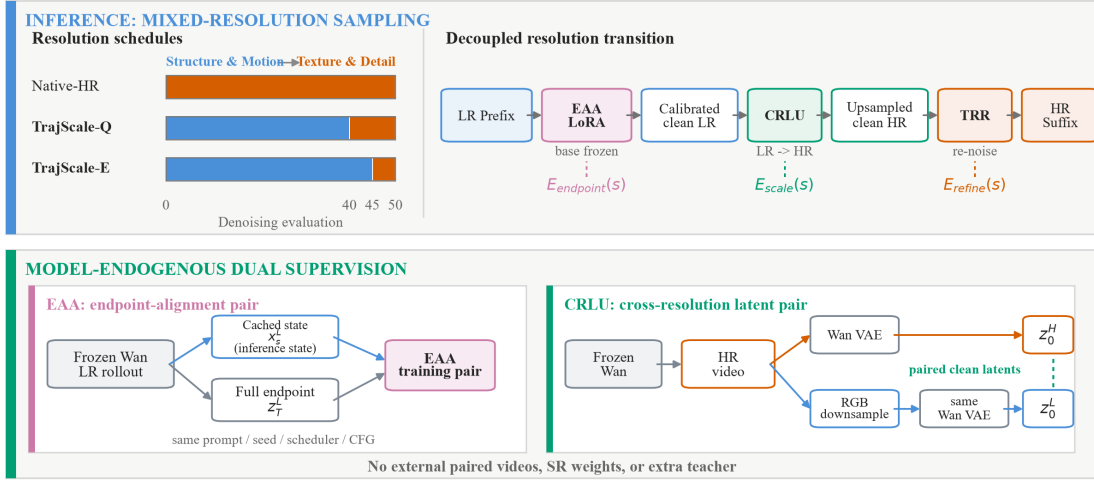


Figure 2: Model-endogenous supervision for EAA and CRLU. EAA pairs a cached transition state with the converged low-resolution endpoint under identical prompt, seed, scheduler, and CFG settings. CRLU pairs clean latents encoded from a frozen-Wan-generated high-resolution video and its RGB-downsampled counterpart. Both supervision streams are generated by the same frozen Wan model.

grid, and Δ_F represents the first-order temporal difference between adjacent latent frames. We set $\lambda_{\text{content}} = 0.2$, $\lambda_{\text{temp}} = 0.1$, and $\epsilon = 10^{-3}$ in our experiments. These three loss terms jointly constrain point-wise reconstruction, low-frequency content preservation, and temporal consistency. Because a low-resolution latent does not uniquely determine high-resolution detail, CRLU learns a stable clean-latent correspondence across scales. The remaining high-frequency content is synthesized during the high-resolution suffix initialized by TRR.

3.4 Handoff-Local Endpoint Alignment

EAA is implemented via LoRA (Hu et al. 2022) and is activated exclusively at the designated transition step. The base Wan model remains completely frozen; low-rank updates of rank r are injected solely into the q, k, v, o projections of the attention modules and the $fn.0, fn.2$ linear layers of the Feed-Forward Networks (FFNs):

$$W' = W + \frac{\alpha}{r} BA.$$

Under the flow-matching parameterization, the EAA-calibrated clean latent prediction at step s is:

$$\tilde{z}_s^L = x_s^L - \sigma_s v_{\phi + \Delta\theta_s}(x_s^L, s, c).$$

The alignment target is the low-resolution trajectory endpoint z_T^L obtained by fully running the frozen model up to step T under identical conditions (text prompt, random seed, initial noise, and scheduler). The endpoint alignment objective is:

$$\mathcal{L}_{\text{EAA}} = \|\tilde{z}_s^L - z_T^L\|_1 + 0.1 \|\tilde{z}_s^L - z_T^L\|_2^2.$$

For earlier transitions (such as step 40), we introduce an additional temporal residual alignment loss with a weight of 0.05 to mitigate the propagation of frame-to-frame errors into the CRLU module. While our framework also supports

direct supervision via the velocity target $v^* = \frac{x_s^L - z_T^L}{\sigma_s}$, all experiments in this paper report results optimized using the clean latent alignment objective above.

By reducing the residual discrepancy between the handoff prediction and the converged low-resolution endpoint, EAA moves the CRLU input toward its clean-latent training domain. TrajScale obtains its system-level acceleration from the low-resolution prefix, while EAA improves state consistency at fixed transition points. Earlier transitions exhibit larger endpoint gaps and therefore leave more room for EAA calibration.

3.5 Cached-Prefix State Consistency

Assuming EAA is activated strictly at the transition step s , its LoRA parameters satisfy $\Delta\theta_k = 0$ for all steps $k < s$. Given a deterministic sampler alongside fixed text embeddings, random seeds, schedulers, and Classifier-Free Guidance (CFG) scales, the adapted generative process shares the identical transition functions as the base process prior to step s . Thus, by induction over the sampling steps, we establish:

$$x_k^{\text{adapted}} = x_k^{\text{base}}, \quad \forall k \leq s.$$

This property guarantees that the pre-cached base model prefix state x_s^L is mathematically identical to the state actually accessed by EAA during inference. For localized adaptations restricted to a single transition step, caching the prefix state allows us to bypass redundant prefix computations entirely during inference without introducing any training-inference distribution drift, maintaining an $O(1)$ state retrieval overhead.

This consistency property relies on four strict conditions: EAA must be engaged exclusively at the transition step; training and inference must share identical schedulers, CFG scales, and conditional inputs; the prefix sampling path must be entirely deterministic; and no unaligned stochastic operations can occur within the prefix trajectory. If EAA were

extended across multiple steps, its parameters would alter subsequent states, requiring advanced training mechanisms such as student model rollouts, synchronized teacher predictions, or Exponential Moving Average (EMA) teacher models.

3.6 Target-Resolution Re-noising and Operating Points

Once EAA produces the endpoint-calibrated low-resolution clean latent, CRLU outputs the upscaled high-resolution clean latent:

$$\hat{z}_s^H = U_\psi^{\text{CRLU}}(\hat{z}_s^L).$$

TRR reconstructs a target-resolution noisy state using the schedule coefficient σ_{s+1} of the next timestep:

$$x_{s+1}^H = (1 - \sigma_{s+1})\hat{z}_s^H + \sigma_{s+1}\epsilon^H.$$

The remaining denoising calculations from step $s + 1$ to T are subsequently completed by the frozen Wan denoiser operating on the high-resolution latent space. Unlike an *Interpolated-Flow Transition* that resizes a low-resolution velocity field, TRR samples target-resolution noise and reconstructs the state at the intended schedule level. The remaining suffix therefore synthesizes newly introduced spatial frequencies with the frozen model’s native high-resolution prior.

Our **50-step pipeline** uses Wan2.1 T2V-1.3B, configured with 50 inference steps, a sample shift of 8, and a CFG scale of 6. Extensive transition-step scanning identified two representative operational points. The quality-centric **TrajScale-Q** (TrajScale@40) reserves 10 high-resolution steps, granting a larger budget for texture generation while managing a wider residual trajectory gap that requires substantial EAA alignment. Conversely, the efficiency-centric **TrajScale-E** (TrajScale@45) reserves only 5 high-resolution steps to minimize latency; its transition state sits closer to a fully converged clean latent, resulting in more stable CRLU inputs and narrower EAA corrections. Together, these two instances define the quality-efficiency Pareto frontier of TrajScale.

Our **4-step pipeline** leverages Wan2.1 T2V-14B StepDistill-CfgDistill, operating on four nominal timesteps [1000, 750, 500, 250] with a sample shift of 5. The distilled instance **TrajScale-D4** (TrajScale@3-of-4) runs low-resolution sampling for the first two steps. At step 3, EAA predicts the endpoint-calibrated low-resolution clean latent, which is upsampled by CRLU and re-noised by TRR for step 4 of the high-resolution trajectory. A single final high-resolution denoising step completes the generation.

4 Experiments

4.1 Experimental Setup

We evaluate TrajScale at both system and module levels, covering end-to-end quality-latency trade-offs, the individual and interactive effects of CRLU and EAA, and compatibility with a 4-step distilled sampler.

For the 50-step pipeline, we generate approximately 1,000 reference videos using Wan2.1 T2V-1.3B. Each video contains 81 frames at 720×1248 resolution and 16 fps. Bicubic

RGB downsampling produces corresponding 480×832 and 368×640 videos, which are re-encoded by the frozen Wan VAE to construct aligned cross-resolution latent pairs. EAA training pairs consist of the low-resolution state immediately preceding a designated transition and the endpoint of the corresponding complete low-resolution trajectory.

The 4-step experiments use approximately 5,000 videos generated by Wan2.1 T2V-14B StepDistill-CfgDistill. CRLU is trained on $368 \times 640 \rightarrow 720 \times 1248$ clean latent pairs, while EAA pairs the state preceding step 3 with the complete step-4 low-resolution endpoint. Training, validation, and test partitions are separated by both text prompts and random seeds.

CRLU is optimized with AdamW for up to 50k steps using a learning rate of 1×10^{-4} , *bf16* precision, and an EMA decay of 0.9999. EAA is trained for up to 10k steps with AdamW, a learning rate of 5×10^{-5} , and LoRA rank/scale 32/32. Training uses four NVIDIA H100 GPUs and requires approximately 8 hours for CRLU and 33 hours for EAA. All efficiency measurements are averaged over five independent cold-start runs. Since all supervision is generated by the frozen target model, TrajScale requires neither external paired videos nor an additional upscaling teacher.

4.2 Baselines and Evaluation Protocol

System-level comparisons include **Native-HR Sampling**, which executes all denoising steps at 720p; **Trilinear Handoff**, which performs mixed-resolution inference using fixed trilinear latent interpolation; and **Endpoint Re-noising**, which completes the low-resolution trajectory, applies CRLU, and performs one low-intensity high-resolution refinement step. We additionally report **Native-LR Sampling** as a low-resolution reference.

Module interactions are evaluated through a 2×2 factorial design that crosses handoff states {Unaligned, EAA-aligned} with resolution operators {Trilinear, CRLU}. The four resulting configurations are Trilinear Handoff, EAA+Trilinear, CRLU-only, and the complete TrajScale framework. Oracle-Aligned CRLU and Joint Trajectory-Scale Mapper (JTSM) are used as diagnostic references; extended results are provided in the Supplementary Material.

For endpoint alignment, we report endpoint latent L_1 , decoded PSNR and LPIPS, temporal latent differences, and sample-wise win rates. CRLU is evaluated using latent L_1 , PSNR, SSIM, LPIPS, and temporal L_1 . End-to-end quality is measured with VBench (Huang et al. 2024) and human evaluation. We define **VBench-5** as the unweighted mean of sub-ject consistency, background consistency, motion smoothness, aesthetic quality, and imaging quality.

All paired configurations share identical prompts, random seeds, initial noise, schedulers, and guidance parameters. Prompt-level score differences are analyzed using 10,000 paired bootstrap iterations for 95% confidence intervals and two-sided exact sign tests. Human evaluation uses randomized, double-blind comparisons over 10 matched prompts, with three independent judges per prompt and prompt-level majority voting. Inter-rater agreement is reported in the Supplementary Material.

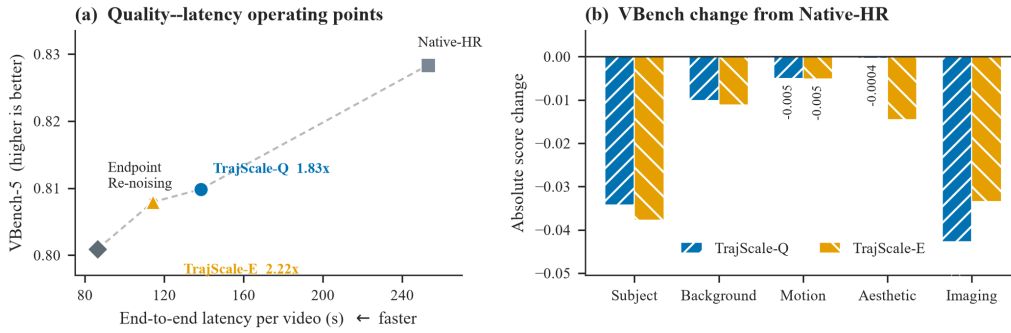


Figure 3: Quality-efficiency trade-offs relative to Native-HR Sampling. (a) TrajScale-Q and TrajScale-E provide quality- and efficiency-oriented operating points, while Endpoint Re-noising achieves higher speed at a larger quality cost. (b) Absolute per-dimension VBench changes relative to Native-HR.

Efficiency measurements include end-to-end cold-start latency, peak GPU memory, low- and high-resolution denoising evaluations, and speedup over Native-HR Sampling. All EAA, CRLU, TRR, VAE, and dynamic model-loading overheads are included.

5 Results and Analysis

5.1 System-Level Quality-Efficiency Trade-off

Table 1 compares the standard 50-step Native-HR pipeline with the two principal TrajScale operating points. TrajScale-Q moves the first 40 evaluations to the low-resolution grid and retains 10 high-resolution suffix evaluations, reducing latency from 253.10 to 138.36 seconds (1.83 $\times$). TrajScale-E retains only five high-resolution evaluations and further reduces latency to 114.26 seconds (2.22 $\times$). Their VBench-5 scores are 0.80983 and 0.80792, compared with 0.82836 for Native-HR Sampling.

The Endpoint Re-noising baseline reaches 86.45 seconds (2.93 $\times$) but decreases VBench-5 to 0.80093, with inferior scores on 9/10 prompts relative to Native-HR ($p = 0.02148$). TrajScale therefore retains a longer high-resolution suffix when additional structural and texture refinement is beneficial, while later transitions prioritize inference speed. Peak memory remains approximately 26 GiB across Native-HR, TrajScale-Q, TrajScale-E, and Endpoint Re-noising, indicating that the primary system benefit of TrajScale is reduced runtime rather than memory consumption.

5.2 Component Analysis

Cross-resolution latent upsampling. Table 2 evaluates CRLU against trilinear interpolation over 50 paired clean latent samples. For 480p $\rightarrow$ 720p, CRLU reduces latent L_1 from 0.1929 to 0.0991, corresponding to a 48.6% reduction. It also improves decoded PSNR by 5.71 dB, reduces LPIPS from 0.2358 to 0.0629, and lowers temporal L_1 by 30.8%. CRLU wins on all 50 samples for PSNR, LPIPS, and temporal consistency, and on 49/50 samples for SSIM.

The more aggressive 368p $\rightarrow$ 720p transition shows the same trend: latent L_1 , LPIPS, and temporal L_1 are reduced by 33.4%, 51.1%, and 15.5%, respectively. These consistent latent, perceptual, and temporal improvements indicate that CRLU learns a substantially more faithful cross-resolution mapping than fixed interpolation.

Endpoint alignment. At an alignment intensity of 0.75, EAA reduces the residual low-resolution endpoint L_1 error by 25.82% at step 40 and 21.03% at step 45. Both settings improve all 10 evaluation samples, with paired confidence intervals excluding zero. EAA also increases decoded PSNR by 2.55 and 2.23 dB and reduces latent temporal difference errors by 23.44% and 21.24%, respectively. On the 4-step distilled model, EAA provides a smaller but consistent 5.03% endpoint-error reduction. All three comparisons have a 10/0 paired win rate and two-sided exact sign-test $p = 0.00195$; full confidence intervals are reported in Supplementary Table S2.

Alignment-intensity validation reveals a non-monotonic dependence on correction strength. An intermediate intensity of 0.75 provides the lowest endpoint error and the strongest downstream quality, whereas weaker or full-strength corrections are less effective. This behavior supports calibrated rather than unconstrained endpoint correction.

Factorial effects. The 2×2 analysis in Table 3 shows that CRLU provides the principal broad quality gain, improving VBench-5 by approximately 0.031–0.041 across the three evaluated pipelines. EAA serves a more targeted role. With CRLU fixed, it improves the imaging-quality dimension by +0.00599, +0.00254, and +0.00479 at step 40, step 45, and on the 4-step sampler, respectively. These detail-oriented improvements are distributed across only one component of the five-dimensional aggregate, and therefore appear as smaller changes in the averaged VBench-5 score.

The results support a clear division of labor. CRLU restores the cross-resolution latent representation and supplies the dominant improvements in motion smoothness, aesthetics, and overall imaging quality. EAA further calibrates the handoff state toward the converged low-resolution endpoint, enriching localized textures and fine structures that may be underrepresented by aggregate automatic metrics.

5.3 Perceptual Effects and Distilled-Sampler Compatibility

Human evaluation confirms that the detail recovery introduced by EAA is perceptually visible. With the same CRLU operator, EAA receives fine-detail win/loss/tie counts of 8/2/0 at step 40 and 9/0/1 at step 45. Overall-quality judgments are also positive, reaching 6/2/2 and 6/0/4, re-

Method	LR/HR evals.	Time (s) ↓	Latency red.	Speedup ↑	VBench-5 ↑	Retention ↑
Native-HR Sampling	0/50	253.10 ± 1.53	—	1.00×	0.82836	100.00%
TrajScale-Q (Step 40)	40/10	138.36 ± 0.50	45.33%	1.83×	0.80983	97.76%
TrajScale-E (Step 45)	45/5	114.26 ± 1.52	54.85%	2.22×	0.80792	97.53%

Table 1: System-level comparison against Native-HR Sampling. Timings average five cold-start runs, and quality is evaluated on ten matched prompts with all processing overheads included.

Input → Output	Operator	Latent L_1 ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Temp. L_1 ↓
480p → 720p	Trilinear	0.1929	24.29	0.7159	0.2358	0.02121
480p → 720p	CRLU	0.0991	30.00	0.8696	0.0629	0.01468
368p → 720p	Trilinear	0.2444	23.42	0.6740	0.3352	0.02388
368p → 720p	CRLU	0.1628	24.10	0.7592	0.1640	0.02017

Table 2: CRLU versus trilinear interpolation on paired clean latents ($n = 50$).

Sampler / transition	Trilinear	EAA+Tri.	CRLU-only	TrajScale	CRLU gain	EAA gain	Total gain
Wan50 / Step 40 (TrajScale-Q)	0.77812	0.77901	0.80896	0.80983	+0.03085	+0.00087	+0.03171
Wan50 / Step 45 (TrajScale-E)	0.76776	0.76734	0.80842	0.80792	+0.04066	-0.00050	+0.04016
Distill4 / Step 3-of-4	0.81796	0.81909	0.85670	0.85680	+0.03874	+0.00010	+0.03884

Table 3: End-to-end factorial results on VBench-5 ($n = 10$). “EAA gain” denotes the marginal effect of enabling EAA with CRLU fixed.

Comparison	Details	Overall	Temporal
EAA vs. Unaligned, CRLU (S40)	8/2/0	6/2/2	1/9/0
EAA vs. Unaligned, CRLU (S45)	9/0/1	6/0/4	0/8/2
CRLU-only vs. Trilinear (S45)	10/0/0	10/0/0	8/0/2
TrajScale-E vs. Trilinear (S45)	10/0/0	10/0/0	9/0/1

Table 4: Prompt-level majority votes (W/L/T) over ten prompts with three raters per prompt. Bold entries have exact sign-test $p < 0.05$.

spectively. These preferences support the dimension-level VBench observation that EAA selectively improves imaging quality and localized structure.

The additional fine-scale content introduces a spatial-temporal trade-off: temporal stability favors the unaligned CRLU state in both settings. This suggests that EAA shifts the operating point toward sharper local rendering, while CRLU and the remaining high-resolution suffix provide the broader spatial and temporal reconstruction.

CRLU provides the broader perceptual improvement: at step 45, both CRLU-only and TrajScale-E obtain 10/10 prompt-level wins over trilinear handoff for fine details and overall quality, while also improving temporal stability. EAA adds a narrower refinement of local detail on top of this cross-resolution reconstruction.

TrajScale is also compatible with the 4-step distilled Wan sampler. On the 368p trajectory, Trilinear Handoff achieves a VBench-5 score of 0.81796, whereas TrajScale-D4 reaches 0.85680. The associated cold-start latency increases only from 167.09 to 170.29 seconds. As in the 50-step setting, CRLU supplies the primary cross-resolution quality improvement, while EAA performs localized endpoint calibration and improves the imaging-quality component.

Qualitative comparisons in Figure 1(a) further illustrate CRLU’s recovery of high-frequency textures and EAA’s localized adjustments. Extended spatial crops and temporal sequences are provided in Figure S1. The content-matched

Native-HR (estimated) reference is produced from a complete 368p trajectory and is used only for frame-level structural comparison; true 720p Native-HR Sampling is used for the quantitative evaluation in Table 1.

6 Discussion and Limitations

TrajScale’s decomposition keeps CRLU in a stable clean-latent domain and delegates underdetermined detail to the frozen high-resolution suffix. Endpoint metrics, imaging-quality gains, and human preferences show that EAA improves localized detail, although its effect depends on the CRLU input distribution and suffix length.

Several limitations remain. CRLU is tied to the target VAE, generator, and scale, and its training inputs differ from native low-resolution trajectory states. Fixed transitions require step-specific EAA parameters, motivating timestep-conditioned adapters. TrajScale also targets in-trajectory spatial scaling rather than real-world degradations, and inherits the base generator’s content risks and safety requirements.

7 Conclusion

TrajScale decouples mixed-resolution sampling into end-point alignment, clean-latent upsampling, and target-resolution re-noising through EAA, CRLU, and TRR. With model-endogenous supervision, its quality- and efficiency-oriented variants achieve 1.83× and 2.22× speedups while retaining 97.76% and 97.53% of Native-HR VBench-5. CRLU consistently outperforms interpolation, while EAA improves endpoint alignment and localized detail across standard and distilled samplers.

References

Blattmann, A.; Rombach, R.; Ling, H.; Dockhorn, T.; Kim, S. W.; Fidler, S.; and Kreis, K. 2023. Align Your Latents: High-Resolution Video Synthesis with Latent Diffu-

sion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Gu, Y.; Fang, G.; Jiang, Y.; Mao, W.; Han, S.; Cai, H.; and Shou, M. Z. 2026. AnyFlow: Any-Step Video Diffusion Model with On-Policy Flow Map Distillation. arXiv:2605.13724.

HaCohen, Y.; Brazowski, B.; Chiprut, N.; Bitterman, Y.; Kvochko, A.; Berkowitz, A.; Shalem, D.; Lifschitz, D.; Moshe, D.; Porat, E.; Richardson, E.; Shiran, G.; Chachy, I.; Chetboun, J.; Finkelson, M.; Kupchick, M.; Zabari, N.; Guetta, N.; Kotler, N.; Bibi, O.; Gordon, O.; Panet, P.; Benita, R.; Armon, S.; Kulikov, V.; Inger, Y.; Shiftan, Y.; Melumian, Z.; and Farbman, Z. 2026. LTX-2: Efficient Joint Audio-Visual Foundation Model. arXiv:2601.03233.

Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*.

Huang, X.; Li, Z.; He, G.; Zhou, M.; and Shechtman, E. 2025. Self Forcing: Bridging the Train-Test Gap in Autoregressive Video Diffusion. arXiv:2506.08009.

Huang, Z.; He, Y.; Yu, J.; Zhang, F.; Si, C.; Jiang, Y.; Zhang, Y.; Wu, T.; Jin, Q.; Chanpaisit, N.; Wang, Y.; Chen, X.; Wang, L.; Lin, D.; Qiao, Y.; and Liu, Z. 2024. VBench: Comprehensive Benchmark Suite for Video Generative Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Jeong, W.; Lee, K.; Seo, H.; and Chun, S. Y. 2025. Training-Free Mixed-Resolution Latent Upsampling for Spatially Accelerated Diffusion Transformers. arXiv:2507.08422.

Luo, S.; Tan, Y.; Huang, L.; Li, J.; and Zhao, H. 2023. Latent Consistency Models: Synthesizing High-Resolution Images with Few-Step Inference. arXiv:2310.04378.

ModelTC. 2026. LightX2V: Variable Resolution Inference and Step Distillation Documentation. <https://github.com/ModelTC/LightX2V>. Accessed: 2026-07-15.

Rombach, R.; Blattmann, A.; Lorenz, D.; Esser, P.; and Ommer, B. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Wan, T.; Wang, A.; Ai, B.; Wen, B.; Mao, C.; Xie, C.-W.; Chen, D.; Yu, F.; Zhao, H.; Yang, J.; Zeng, J.; Wang, J.; Zhang, J.; Zhou, J.; Wang, J.; Chen, J.; Zhu, K.; Zhao, K.; Yan, K.; Huang, L.; Feng, M.; Zhang, N.; Li, P.; Wu, P.; Chu, R.; Feng, R.; Zhang, S.; Sun, S.; Fang, T.; Wang, T.; Gui, T.; Weng, T.; Shen, T.; Lin, W.; Wang, W.; Wang, W.; Zhou, W.; Wang, W.; Shen, W.; Yu, W.; Shi, X.; Huang, X.; Xu, X.; Kou, Y.; Lv, Y.; Li, Y.; Liu, Y.; Wang, Y.; Zhang, Y.; Huang, Y.; Li, Y.; Wu, Y.; Liu, Y.; Pan, Y.; Zheng, Y.; Hong, Y.; Shi, Y.; Feng, Y.; Jiang, Z.; Han, Z.; Wu, Z.-F.; and Liu, Z. 2025. Wan: Open and Advanced Large-Scale Video Generative Models. arXiv:2503.20314.

Xu, Y.; Park, T.; Zhang, R.; Zhou, Y.; Shechtman, E.; Liu, F.; Huang, J.-B.; and Liu, D. 2024. VideoGigaGAN: Towards Detail-Rich Video Super-Resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Yin, T.; Gharbi, M.; Park, T.; Zhang, R.; Shechtman, E.; Durand, F.; and Freeman, W. T. 2024. Improved Distribution Matching Distillation for Fast Image Synthesis. arXiv:2405.14867.

Zhai, Y.; Lin, K.; Yang, Z.; Li, L.; Wang, J.; Lin, C.-C.; Doermann, D.; Yuan, J.; and Wang, L. 2024. Motion Consistency Model: Accelerating Video Diffusion with Disentangled Motion-Appearance Distillation. arXiv:2406.06890.

Zhao, C.; Chen, J.; Li, H.; Kang, Z.; Lu, S.; Wei, X.; Zhang, K.; Yang, J.; and Tai, Y. 2026. LUVE: Latent-Cascaded Ultra-High-Resolution Video Generation with Dual Frequency Experts. arXiv:2602.11564.

Zheng, X.; Liu, X.; Ding, Y.; Feng, W.; Lin, J.; Guo, J.; and Qin, H. 2026. Multi-Resolution Flow Matching: Training-Free Diffusion Acceleration via Staged Sampling. arXiv:2607.01642.

Zhou, S.; Yang, P.; Wang, J.; Luo, Y.; and Loy, C. C. 2024. Upscale-A-Video: Temporal-Consistent Diffusion Model for Real-World Video Super-Resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.